

STAR Method

1. Tell me about a time you encountered a difficult problem and how you went about solving it?

I worked on a project regarding predicting viewers' sightlines in 360 videos. A group member got sick and couldn't continue. We could have finished the project without their components, but because I had finished my responsibilities ahead of schedule and their part was closely related to my own, I volunteered to take on the work.

My initial role in the project was predicting which tile the viewers would be looking at based on sudden sounds. My new responsibility was to predict using unique sounds like human dialogue or a dog barking.

I already had the components for 4 channel audio mapping onto a sphere figured out. Instead of tackling this alone, I got strong insight from my teammates' advice. This particular teammate was working on standout visuals. After some research, I decided to focus on filtering for frequencies between 2000 and 5000hz. My goal was to locate sounds the human ear fixated on.

We were able to complete the project on time with every planned component. Testing individual components we concluded that with the addition of both surprising audio and unique audio filtering, we were able to get a stronger saliency score on videos that have a singular strong focal point.

What I took away from this experience was that being diligent with my task gave me the flexibility to take ownership when something unexpected happened.

2. Tell me about a time you disagreed with a teammate or had a conflict within a group. How did you handle it?

During a group project my teammate and I were at odds on one part of the implementation. I thought my approach was simpler and effective, while my teammate preferred a different approach.

We needed to decide on an approach without letting our conflicts slow down the project. I had to find a solution that allowed us to work effectively.

Rather than arguing or splitting our resources, I asked them to explain their thoughts and I was able to gain insight on what their concerns were. I touched on parts I agreed with as to not alienate them before I also explained my thoughts and concerns.

We realized our main conflict came from having different views on what the requirements were for the project. We successfully finished the assignment after we managed to decide on an approach that we were both comfortable with.

3. Tell me about a time you had to complete something under a tight deadline. How did you manage it?

During my cyber security internship, there was an ongoing attack. An alert came through our SIEM showing an employee logged in from an unusual location and started accessing files.

The account had low access and considering the threat level my mentor decided to let me handle it as he was tied up with something already. He wanted me to give him an answer within 30 minutes.

Because of the time constraints I knew I couldn't investigate everything. I decided to figure out what the user has done so far. Which account they managed to log onto. All while documenting it in an open case. The account was attempting to send mass emails to other contacts within the company's address.

Before the 30 minutes elapsed I recommended we alert everyone to not click on links from that email address and to lock the compromised account. My mentor went ahead and locked their account and blocked access to that webpage as well as alerting the other departments about the rogue emails.

4. Tell me about a time you had to learn something new or unfamiliar in order to complete a task.

I was working on a project that analyzed music and automatically generated useful tags such as genre and BPM, then uploaded that information to a database. Although I had learned a lot of database theory, I had never actually implemented a database myself, and I also had no prior experience with audio processing.

My goal was to learn enough about both areas to create a program that could identify a song's title and artist, analyze the audio to determine things like BPM and genre, and then upload that information to a database. The database also needed to support multiple artists while still identifying a primary artist.

I started by breaking the problem into smaller tasks so I could research and learn each part systematically. First, I researched how audio files could be broken down into useful characteristics. I learned how BPM could be detected and how different elements of a song, such as bass and drums, could be used as indicators when classifying genres. I then experimented with different audio-processing techniques and tested them against songs where I already knew the BPM and genre to see how accurate my results were.

Once I understood the audio-processing side, I moved on to the database. I applied what I had learned from database theory to design the tables and relationships, including a separate relationship for songs and their multiple artists while still being able to identify the main artist. I built the database incrementally and tested each part before connecting it to the audio-processing program.

As a result, I was able to take a song as an input, extract information from it, generate the appropriate metadata, and store that information in a structured database. More importantly, I learned that when I encounter technology I'm unfamiliar with, I don't need to understand the entire subject at once. Breaking the problem into smaller pieces and learning only what I need for each step allowed me to make steady progress and complete the project.

5. Tell me about a time you noticed something that needed to be done and took action without being asked.

During a group project, one of my teammates became sick and started having difficulty completing their portion of the assignment. I noticed early on that they were struggling, so before the roles and responsibilities were finalized, I suggested that they take on a part of the project that was closely related to mine. That way, we could easily collaborate and I could help if they needed it.

As their condition got worse, they eventually became too tired to continue working on their portion. Although nobody asked me to, I proactively stepped in and took over the programming they were unable to complete. I also made sure to keep communicating with them throughout the process so they could still contribute their ideas and stay involved with the project.

As a result, we completed the project without delaying the rest of the team, and my teammate was still able to feel proud of their contribution. The experience taught me that being proactive isn't just about taking on extra work; it's also about recognizing when someone needs support and taking action before the problem affects the entire team.

6. Tell me about a particularly difficult technical problem you encountered. How did you approach solving it?

I was working on a 360-degree video gaze prediction model and noticed our model took a long time to process a video. It was going to cause issues down the road if we wanted to bulk process large quantities of videos, and it also made testing the program arduous.

Before we wasted a large amount of development time waiting around for the model to finish, I wanted to find a way to speed the system up.

After investigating I dialed in on how each of us are in charge of building a component. My teammates and I each made components that produced a 360-degree video tile saliency map. It assigned weights to tiles on every frame. My first solution was to test on fewer frames by only processing every other frame. However, I also noticed that each of our visual components independently extracted the same frames, while each of our audio components independently mapped the sound to a direction. I suggested that we preprocess the video and audio once before running them through our individual components, and collaborated with my teammates to implement that approach.

The videos were able to be processed by every component much faster. We reduced the final processing time to roughly one-third of what it had been, and testing times were almost cut in half as well.

7. Tell me about a time you received criticism or feedback that you disagreed with. How did you respond?

During a cybersecurity internship, I was working on a task where I had to document some security findings and explain their potential impact. After reviewing my work, my supervisor told me that my documentation was too detailed and that I should focus more on the most important findings rather than explaining every technical detail.

Initially, I disagreed because I felt that including more technical information would make the documentation more useful, especially for someone who wanted to understand how I reached my conclusions.

Instead of immediately pushing back, I asked my supervisor what they felt the documentation should accomplish and who the primary audience was. I realized that the report was primarily intended for people who needed to quickly understand the risks and decide what actions to take, rather than reproduce my technical analysis. I went back through the report and separated the key findings and recommendations from the supporting technical details. I also kept the more detailed information available as supporting documentation.

The revised version was much easier to review, and my supervisor was satisfied with it. More importantly, I learned that good technical communication isn't necessarily about including as much information as possible. It's about giving the right information to the right audience. Since then, I've tried to understand the purpose and audience of my work before deciding how much detail to include.

8. Tell me about a time when you had multiple responsibilities competing for your attention. How did you decide what to prioritize?

I was in my final semester of college when my dog suddenly became severely sick due to his trachea collapsing. I was the only family member who was able to take him to his veterinary appointments, while at the same time I had final assignments and projects that I needed to complete.

I realized that I couldn't simply try to brute force this situation, because it was time sensitive and unstable, I prioritized my dog's immediate health. I communicated my situation to my teammates and professors, and I asked my professors if I could receive some course material in advance so I could work ahead whenever my dog's condition was stable. I also enlisted the help of my siblings, and we made plans for one of them to temporarily move in with me and help with his care. These steps allowed me to take some emotional weight off of myself, preserving enough time and energy for me to effectively finish my degree.

Whenever my dog was stable, I used that time to focus heavily on my coursework. By communicating early, planning ahead, and getting help from my family, I was able to balance his care with my academic responsibilities.

I successfully graduated that semester. Unfortunately, my dog passed away shortly afterward, but I was grateful that I had been able to be there for him while still finishing my degree. The experience taught me a lot about resilience and showed me that when I have multiple responsibilities competing for my attention, I need to prioritize what is most urgent, communicate early, and ask for help rather than trying to handle everything alone.

9. Tell me about a time you helped a teammate who was struggling with something. What did you do?

During a group project in one of my programming classes, one of my teammates was struggling with part of the implementation. They understood what the program was supposed to do, but they were having trouble translating the requirements into working code.

We needed to get that portion of the project working while making sure my teammate actually understood the solution, rather than me simply taking over their part.

I first asked them to walk me through what they had tried and where they were getting stuck. That helped me realize that the main issue wasn't their coding ability, but that they were approaching the problem in a way that made it more complicated than it needed to be. I broke the problem down into smaller steps and worked through one example with them. Rather than writing the code for them, I had them implement each step while I answered questions and pointed out issues when they came up. Once they had a working version, we tested it together with a few different inputs.

They were able to finish their portion of the project and explain how the solution worked. We were able to integrate their work with the rest of the project and finish the assignment on time. It also reinforced for me that helping a teammate isn't always about giving them the answer. Sometimes it's more effective to help them understand the problem so they can solve it themselves.

10. Tell me about a time you made a mistake or failed at something. What did you do afterward?

During a project, I was working on a part of the implementation that involved technology I didn't have much experience with. I initially assumed I could figure it out on my own, so I spent too much time trying different approaches and troubleshooting without making much progress.

I was responsible for getting my portion completed, but I eventually realized that I was falling behind and that continuing to work on it alone wasn't a good use of my time.

I took a step back and recognized that the problem wasn't just that I hadn't found the right solution yet. I didn't have enough familiarity with the technology to efficiently solve the issue on my own. Rather than continuing to spin my wheels, I reached out to someone with more experience and explained what I had tried, where I was stuck, and what I didn't understand. Their guidance helped me identify where my approach was going wrong. I then went back and completed the work with a much better understanding of what I was doing.

I was able to finish my portion, but I wouldn't describe the experience as a major success. I had spent more time struggling than I needed to because I waited too long to ask for help. The bigger takeaway was becoming more aware of my own limitations. Now, when I'm working on something unfamiliar, I give myself time to investigate independently, but I also pay attention to when I'm no longer making meaningful progress. If I reach that point, I ask for help sooner rather than letting my pride or reluctance to bother someone turn a small problem into a larger one.